

Git Merge Conflict HOL Report

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Step 1: Verify clean state

git status

```
PS G:\My Drive\JAVA-FSE> git status
On branch main
Your branch is up to date with 'origin/main'.
```

Step 2: Create branch GitWork

```
nothing to commit, working tree clean
PS G:\My Drive\JAVA-FSE> git branch GitWork
PS G:\My Drive\JAVA-FSE> git checkout GitWork
Switched to branch 'GitWork'
```

Step 3: Create hello.xml

```
PS G:\My Drive\JAVA-FSE> echo "<hello>Branch Version</hello>" > hello.xml
PS G:\My Drive\JAVA-FSE> git status
On branch GitWork
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.xml

nothing added to commit but untracked files present (use "git add" to track)
PS G:\My Drive\JAVA-FSE>
```

Step 4: Commit branch changes

```
git add .
git commit -m "Add hello.xml in GitWork"
```

```
PS G:\My Drive\JAVA-FSE> git add .
PS G:\My Drive\JAVA-FSE> git commit -m "Add hello.xml in GitWork"
[GitWork a71aa0f] Add hello.xml in GitWork
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 hello.xml
PS G:\My Drive\JAVA-FSE>
```

Step 5: Switch to main

git checkout main

```
PS G:\My Drive\JAVA-FSE> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS G:\My Drive\JAVA-FSE>
```

Step 6: Modify hello.xml on main

echo "<hello>Main</hello>" > hello.xml

```
PS G:\My Drive\JAVA-FSE> echo "<hello>Main Version</hello>" > hello.xml
```

Step 7: Commit on main

```
git add .
git commit -m "Update hello.xml on main"
```

```
PS G:\My Drive\JAVA-FSE> git add .
PS G:\My Drive\JAVA-FSE> git commit -m "Add hello.xml in main"
[main a33effb] Add hello.xml in main
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 hello.xml
PS G:\My Drive\JAVA-FSE>
```

Step 8: View log

git log --oneline --graph --decorate --all

```

PS G:\My Drive\JAVA-FSE> git log --oneline --graph --decorate --all
* a33effb (HEAD -> main) Add hello.xml in main
| * a71aa0f (GitWork) Add hello.xml in GitWork
|/
* d5b28e2 (origin/main) Add Git Branching and Merging HOL report
* 9102cee Add sample file in GitNewBranch
* 6e77c7d Add Git Ignore HOL report
* 988896c Implement .gitignore for log files
* f67aea7 Added Git HOL Commands and Report PDF
* f3fc4fb Remove Git HOL Commands and Report PDF
* 826fcc5 Add Week 6 Git Ignore hands-on exercises
* f25e3e5 Add Week 5 React hands-on exercises
* c3fbd07 Add README.md for Week 5 exercises and objectives
* 4891bb7 Add Week 5 React files
* 9fe31b2 Add Week 4 Microservices with Spring Boot 3 and Spring Cloud
* f1d6712 Add source directory README
* 4a7c0b7 Add Week 3 Spring Core, Maven, Spring Boot and Hibernate modules
* 3240c51 Update Week 1 README
* 23ca687 Add Week 2 exercises and documentation
* 5e5002e Add Week 1 exercises and documentation
* f66c817 Add README for Java Full Stack Engineer program
* 1fba707 Enhance project documentation and update README
* 4860e9c Update Singleton Pattern
* 3848120 Add Week 1 Design Pattern implementations
* 08bda1e Refactor exercises, reorganize folders, and update PL/SQL and DSA files
* 63be4f1 Add gitignore
* bd68d50 Add PL SQL exercises
* d9cd130 Add DSA exercises
PS G:\My Drive\JAVA-FSE>

```

Step 9: Compare differences

git diff main GitWork

```

PS G:\My Drive\JAVA-FSE> git diff main GitWork
diff --git a/hello.xml b/hello.xml
index 82ff198..674cc52 100644
Binary files a/hello.xml and b/hello.xml differ
PS G:\My Drive\JAVA-FSE>

```

Step 10: Visual diff (P4Merge)

git difftool main GitWork

```
PS G:\My Drive\JAVA-FSE> git difftool main GitWork
```

```
This message is displayed because 'diff.tool' is not configured.  
See 'git difftool --tool-help' or 'git help config' for more details.  
'git difftool' will now attempt to use one of the following tools:  
kmpare emerge vimdiff nvimdiff
```

```
Viewing (1/1): 'hello.xml'  
Launch 'vimdiff' [Y/n]?
```

Step 11: Merge branch

```
git merge GitWork
```

```
PS G:\My Drive\JAVA-FSE> git merge GitWork  
warning: Cannot merge binary files: hello.xml (HEAD vs. GitWork)  
Auto-merging hello.xml  
CONFLICT (add/add): Merge conflict in hello.xml  
Automatic merge failed; fix conflicts and then commit the result.
```

Step 12: Resolve conflict

Edit hello.xml and save resolved version

```
<<<<<< HEAD  
<hello>Main Version</hello>  
=====  
<hello>Branch Version</hello>  
>>>>>> GitWork
```

Step 13: Commit resolved merge

```
git add hello.xml  
git commit -m "Resolve merge conflict"
```

```
<hello>Main and Branch Merged</hello>
```

Step 14: Ignore backup files

```
echo "*.orig" >> .gitignore  
git add .gitignore  
git commit -m "Update .gitignore"
```

```
PS G:\My Drive\JAVA-FSE> git add hello.xml
PS G:\My Drive\JAVA-FSE> git commit -m "Resolve merge conflict"
[main 0327f00] Resolve merge conflict
```

Step 15: List branches

```
git branch -a
```

```
PS G:\My Drive\JAVA-FSE> echo "*.orig" >> .gitignore
PS G:\My Drive\JAVA-FSE> git add .gitignore
PS G:\My Drive\JAVA-FSE> git commit -m "Update .gitignore"
[main 0c556e6] Update .gitignore
1 file changed, 0 insertions(+), 0 deletions(-)
PS G:\My Drive\JAVA-FSE>
```

Step 16: Delete merged branch

```
git branch -d GitWork
```

```
PS G:\My Drive\JAVA-FSE> git branch -a
GitWork
* main
remotes/origin/main
PS G:\My Drive\JAVA-FSE>
```

Step 17: View final log

```
git log --oneline --graph --decorate
```

```
PS G:\My Drive\JAVA-FSE> git branch -d GitWork
Deleted branch GitWork (was a71aa0f).
```

Step 18: Push changes

```
git push origin main
```



```
PS G:\My Drive\JAVA-FSE> git push origin main
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 12 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (10/10), 1.09 KiB | 139.00 KiB/s, done.
Total 10 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 1 local object
To https://github.com/anuragtiwari-stacks/JAVA-FSE.git
    d5b28e2..0c556e6  main -> main
PS G:\My Drive\JAVA-FSE>
```